Two faces of the rare-earth–iron exchange: two-dimensional terahertz spectroscopy of $TmFeO_3$ as vertex spectroscopy of a dressed singlet ion

Working draft[1]

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One-dimensional spectroscopy measures the *spectrum* of a Hamiltonian; two-dimensional coherent spectroscopy (2DCS) measures its *vertices*—the individual nonlinear terms that convert one coherence into another. We show that in $TmFeO_3$ this correspondence closes completely, because a single symmetry principle organises the entire coupling space. The Tm^{3+} crystal-field levels are non-Kramers singlets: in a real singlet basis time reversal makes every angular-momentum matrix element imaginary, so the iron exchange field, acting *linearly*, reaches only the time-odd quadratures of the crystal-field coherences—and that coupling fails on the data threefold. First-order exchange coupling to the crystal-field spectrum is symmetry-blocked, and everything measured across the two subsystems is a *second-order dressing* with exactly two faces: the exchange field squared, which modulates the crystal-field energy (a single tensor W that performs every coherence conversion but never radiates), and the exchange field paired with the light field, which builds an order-induced electric dipole $g_d(\langle F_x\rangle+\delta F_x)\lambda^2$ (a single constant that drives, radiates, and emits internal magnon–crystal-field sum- and difference-frequency tones). Driven by the measured pulse with measured linewidths and dipole weights, these two objects plus the bare subsystems and one self-absorption filter reproduce the polarisation-resolved 2DCS atlas, with every peak assigned, every structural absence a measurement, and every component validated by a leave-one-out necessity audit. Because $\partial^2 E_{\mathrm{CEF}}/\partial S\,\partial S$ is the same coupling surface whose thermal average drives the spin-reorientation transition, the atlas constitutes a coherent, mode-resolved, real-time measurement of the reorientation mechanism itself, and the entire dressed sector must collapse together at the transition.

I. INTRODUCTION

Linear response cannot distinguish two Hamiltonians with the same normal modes: absorption measures pole positions and dipole weights, and any coupling that merely hybridises modes is absorbed into the mode basis. What it can never see is a *conversion*: a coherence prepared on one transition and re-emitted on another. That is precisely what a cross peak in 2DCS certifies [1, 2], and a conversion requires a specific anharmonic vertex. A cross peak at (ω_τ, ω_t) is therefore not a feature to be fitted but a *measurement of one coupling term*: the delay axis names the stored coherence, the detection axis the emitter, and the signed side constrains the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—a polarisation-resolved 2D atlas can be read as the Hamiltonian itself.

The rare-earth orthoferrites are the natural target for this program. Two qualitatively different subsystems share one THz octave: below the spin reorientation the Fe sublattice of $TmFeO_3$ orders in Γ_2 (F_x, C_y, G_z) [3–5], with quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\mathrm{FM}} = 0.38$ and $q_{\mathrm{AFM}} = 0.90\,\mathrm{THz}$, while the three lowest Tm^{3+} crystal-field (CEF) singlets form a qutrit with $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20\,\mathrm{THz}$. The rare-earth–iron coupling in this family is not a spectroscopic curiosity: it is the engine of the spin-reorientation transition, in which the temperature-dependent occupation of the rare-earth levels rotates the Fe anisotropy, and it is the handle by which THz pulses have been used to steer the Fe order coherently [7, 8]. Yet that coupling has essentially always been treated thermodynamically—as a free-energy surface—or as an empirical anisotropy modulation. A single-cycle THz pulse covers all five modes of both subsystems at once, and measurements in the cross- and same-polarised channels of two field geometries return distinct, reproducible peak inventories. This paper shows that the inventories can be read, term by term, as the coherent, mode-resolved anatomy of the rare-earth–iron exchange—and that one symmetry principle fixes what can appear.

II. THE DRESSING PRINCIPLE

The organising fact is that the Tm^{3+} CEF levels are nondegenerate singlets of a non-Kramers ion. Nondegenerate eigenstates of a real crystal-field Hamiltonian may be chosen real; in that basis every matrix element of the angular momentum $\boldsymbol{J}$ is purely imaginary and antisymmetric. Projected onto the qutrit, $\boldsymbol{J}$ therefore has support *only* on the three time-odd Gell-Mann coordinates $\lambda^{2,5,7}$—the imaginary parts of the three coherences—and the ion has rigorously no permanent moment. Two consequences follow immediately.

First-order exchange is blocked. The Fe sublattice exerts an exchange field $\boldsymbol{h}_{\mathrm{ex}}(\boldsymbol{S})$ on the ion; acting linearly it produces the bilinear $K^-\,\boldsymbol{S}\cdot(\text{projected }\boldsymbol{J})$, which can only touch $\lambda^{2,5,7}$. The data exclude this coupling threefold: linear in $\boldsymbol{S}$ it hybridises modes but cannot convert (no cross peak), its channels cancel over the four-sublattice orbit in Γ_1 combinations to machine precision, and its one allowed component reorients the ground state. Its most direct fingerprint is negative: a static mixer drags the dressed E_{12} line with it, and the measured 2D transfer row sits at the 1D absorption frequency *exactly*, at every tested coupling strength (with the bare splitting recalibrated at each strength so the comparison is fair). The singlet ion is protected from its magnetic environment at first order—the same protection that makes non-Kramers ions poor qubits makes them clean spectroscopic reporters.

Everything measured is second-order dressing, and there are exactly two faces. If the environment cannot enter linearly, the leading surviving objects are quadratic, and the qutrit's two $1 \leftrightarrow 2$ quadratures each admit exactly one:

$$H_W = \boldsymbol{S}^T W \, \boldsymbol{S} \, \lambda^1, \qquad H_g = -\, g_d \, E_z \, (F_x \, \lambda^2). \qquad (1)$$

The spin pair SS is $\mathcal{T}$-even and lands on the $\mathcal{T}$-even quadrature λ^1: this is the *energy face*—the exchange modulation of the crystal field, $W \sim \partial^2 E_{\mathrm{CEF}}/\partial S \, \partial S$. In Γ_2 symmetry leaves two components, $W_1^{yy} S_y^2 \lambda^1$ and $W_1^{xz} S_x S_z \lambda^1$, and the data are consistent with a single scale $W_1^{yy} = W_1^{xz} = W$; the leave-one-out runs below show the two components sharing every conversion task rather than owning separate peaks—one tensor, not two mechanisms. Having no field leg, the energy face never radiates: it is the converter. The light–spin pair $E_z S$ is the *moment face*: with the spin leg condensed on the Γ_2 order parameter it is an order-induced electric dipole of the $1 \leftrightarrow 2$ transition, $d_z^{\mathrm{eff}} = g_d(\langle F_x \rangle + \delta F_x)\lambda^2$, reaching the $\mathcal{T}$-odd quadrature through the $\mathcal{T}$-odd condensed spin. It is the one legal way an electric field talks to λ^2, and one constant serves three ways: it *drives* (the $E \parallel c$ writing of the E_{12} coherence), it *radiates linearly* (the $(q_{\mathrm{AFM}}, E_{12})$ readout), and its retained fluctuation $g_d \, \delta F_x \lambda^2$ is a bilinear emitter that radiates internal sum- and difference-frequency tones at $q_{\mathrm{AFM}} \pm E_{12}$—magnon–crystal-field SFG and DFG, the non-degenerate siblings of on-site second-harmonic generation.

This is the entire cross-subsystem sector. Everything else in the atlas is bare structure: the Fe magnon pair coupling (anisotropy plus Dzyaloshinskii–Moriya, entirely within the Fe sector), the bare driven qutrit with its measured dipoles and its on-site hyperpolarisability $\beta\lambda^1\lambda^2$ (the SHG member of the same quadratic-moment family, permitted because the Tm $4c$ site lacks inversion; Fe at centrosymmetric $4b$ is excluded from the whole family), and one propagation filter. And the energy face is not a new object in orthoferrite physics: $\partial^2 E_{\mathrm{CEF}}/\partial S \, \partial S$ is the coupling surface whose thermal average—through the temperature-dependent rare-earth level occupations—rotates the Fe anisotropy and drives the spin-reorientation transition [3, 4, 8]. The 2DCS atlas measures the same surface coherently: mode-resolved, phase-resolved, in both directions, on picosecond timescales.

III. MODEL, AND WHERE IT IS EXACT

The Fe^{3+} ($S = 5/2$) sublattice is classical, with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya) reproducing q_{FM}, q_{AFM} and Γ_2. Each Tm^{3+} is truncated to its three lowest singlets and carried as the Gell-Mann vector $\lambda^a = \langle \Lambda^a \rangle$. Because every single-ion term is linear in the Λ^a, the classical $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of the driven, damped qutrit, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [SM, Sec. S1]. The Fe–Tm sector is built by projecting $\boldsymbol{S}^T A \, \boldsymbol{J}$ into the qutrit in a transported gauge over the four Tm sublattices [SM, Secs. S2–S3]; the vertex catalogue and the exclusions behind Eq. (1) are exhaustive over that space [SM, Sec. S4]. All spectra use the measured pump (spectrum peaked at 0.53 THz, no DC), measured linewidths as damping rates, and measured oscillator strengths as emission weights. Peaks are identified blindly as 2D local maxima with prominence ≥ 1.5.

IV. SYMMETRY ASSIGNS EACH POLARISATION ONE BLOCK

The Tm $4c$ site carries the mirror m ($z \to -z$), under which $d_{x,y}$ are even and d_z odd, while m_z is even and $m_{x,y}$ odd. Together with the time-reversal statement of Sec. II this fixes the entire optical interface. The projected moment matrix —an output of the PJP construction, not an input—has λ^2 carrying only μ_z and $\lambda^{5,7}$ only $\mu_{x,y}$, which forces $|1\rangle, |2\rangle$ to share a parity and $|3\rangle$ to be opposite:

transition	magnetic	electric	owned by
$1 \leftrightarrow 2$	m_z (λ^2)	d_x (λ^1)	$(E \parallel a, H \parallel c)$
$1 \leftrightarrow 3$	m_x (λ^5)	d_z (λ^4)	$(E \parallel c, H \parallel a)$
$2 \leftrightarrow 3$	m_x (λ^7)	d_z (λ^6)	$(E \parallel c, H \parallel a)$

Mirror parity would additionally permit the mirror-even d_x a diagonal (permanent) part on $\lambda^{3,8}$; a permanent moment does not radiate at any transition frequency, and the detection operators exclude the population generators throughout (a measured rectified line at $\omega_t \to 0$ in an $(E \parallel a, H \parallel c)$ channel would falsify this exclusion). An analyser selects a polarisation *state*, hence both the magnetic and the electric dipole radiating into it; there are only *two* detection operators in the atlas, and the two channels sharing one are constrained to share it exactly (Table I). Reciprocity—one operator per polarisation, driving and radiating alike—then implies that *cross-polarised detection reads exactly the block the pulse did not drive.* This is the structural reason cross-polarisation isolates conversion, and it carries two independent checks: the geometry-B subalgebra closure, where the c-directed drive touches only μ_{12}^z so that λ^{4-7} vanish identically (verified: exactly 0, against $\lambda^{1,2,3} \sim 2.6 \times 10^{-2}$); and in geometry A $F_z \equiv 0$ to 10^{-15}, so the Fe sublattice is radiatively silent in that cross channel and anything seen there at a magnon frequency must be Tm.

Because λ^1 and λ^2 are quadratures of the same coherence, their magnitude spectra coincide to 10%: in *detection* the atlas is insensitive to the $E1/M1$ balance (varying w_{E1} over a factor of 50 moves every geometry-A cross amplitude by at most 0.04 once β is refitted). It is the *drive* that fixes it: the electrically written coherence feeds the geometry-B transfer linearly in w_{E1}, and the measured ratio returns $w_{E1} = 0.54$ (in units of μ). The construction is strictly reciprocal at every level the data can test: operator content and within-family dipole ratios are shared exactly between drive and readout, as is the propagation filter below. Three cross-family scales remain as single calibrated numbers used consistently on both sides: w_{E1} (measured); g_d (one parameter with a drive avatar, a linear-emission avatar, and a condensation-ratio correction $s = 0.42$ on its bilinear piece, the trajectory value overestimating it by about two—the model's canting-angle uncertainty); and the Fe:Tm emission efficiency $c_{\rm Fe}$, bounded from *above* by the non-observation of the $(q_{\rm AFM}, q_{\rm AFM})$ magnon diagonal—a null measurement that is itself the calibration. The last ingredient is propagation: with $f = 8.7$ the sample is optically thick at the E_{13} line (depth $\simeq 6$) and mildly at E_{12} (depth $\simeq 1$), and the same transmission filters the incoming field and the emitted $E1$ light—magnetic-dipole radiation passes freely. This one filter, applied identically both ways, is itself reciprocal.

V. THE PATHWAYS, PEAK BY PEAK

We now give every feature of Fig. 1: what makes it possible, how the excitation reaches the emitter, and what would falsify the assignment. Throughout, the delay axis carries a coherence that *never radiates*—it need only be excitable and able to store phase—while the detection axis is constrained by the emitting multipole. That asymmetry does most of the work.

TABLE I. What each channel detects. An analyser setting selects a polarisation *state*; there are therefore only two detection operators. $w_{E1} = 0.54$ is the $E1/M1$ balance, fixed by the geometry-B transfer (see text); no population generator (λ^3, λ^8) appears in either operator, and the $1 \leftrightarrow 3$ transition is fully dark ($\mu_{13} = 0$ in both polarisations, $d_z(1{\to}3) \leq 0.006$).

$(E \parallel a, H \parallel c)$:	$F_z + \mu\lambda^2 + w_{E1}\lambda^1 + \beta\lambda^1\lambda^2$
A cross	$F_z \equiv 0$: Tm only. $\lambda^{1,2}$ resonant at E_{12}
B same	F_z alive but 50× smaller than the Tm terms
$(E \parallel c, H \parallel a)$:	$F_x + \mu_x\lambda^7 + w'_{E1}\lambda^6 + g_d(\langle F_x\rangle + \delta F_x)\lambda^2$
A same	CEF λ^6; $d_z^{\rm eff}$ (static + dynamical); F_x bounded
B cross	F_x; Tm part $\equiv 0$ by closure

A. Geometry A, cross-polarised

$(q_{\rm AFM}, E_{12}) = 1.00$*: the energy face, read forward; measures* W. The $H \parallel a$ Zeeman field launches the $q_{\rm AFM}$ magnon, which precesses at 0.90 THz through the delay. It cannot radiate here—it has no c-axis moment—but it need not: it modulates the Tm crystal field through $W_1^{yy} S_y^2 \lambda^1$. Each pulse supplies one magnon leg, so the $M_{\rm NL}$ subtraction keeps the cross term, which rectifies,

$$h^1_{\rm NL} = W_1^{yy} A^2 \big[\cos q_{\rm AFM}\tau - \cos q_{\rm AFM}(2t+\tau)\big]\theta(t), \quad (2)$$

the difference piece being a step switched on at the probe that carries the delay phase but no t dependence. Fed into the E_{12} oscillator it gives $M_{\rm NL} \propto \cos(q_{\rm AFM}\tau)[1 - \cos E_{12}t]$: a factorised, rank-one peak. *Why it is possible at all*: an E1-dark magnon is made visible by borrowing the Tm dipole. *Where the energy comes from*: δS_y^2 contains only DC and $2q_{\rm AFM}$, so between free modes this vertex could never emit at ω_{12}; the quantum is drawn from the pulse edge, an intra-pulse difference-frequency process. This is verified rather than asserted—stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades, while switching off the direct Tm drive changes it by $< 10^{-4}$. *Signature*: $\propto A^2_{\rm Fe}$, temperature-flat below ~ 10 K, CEF linewidth along ω_t.

$(E_{12}, 2E_{12}) = 0.96$*: second-harmonic emission; measures* β. The pump stores E_{12}, setting the delay axis at 0.50; the detection axis is its second harmonic at $2E_{12} = 1.0000$ THz. A moment linear in the qutrit coordinates cannot radiate a harmonic; the $4c$ site's lack of inversion permits the quadratic term $m_z = \mu\lambda^2 + \beta\lambda^1\lambda^2$, and Fe at $4b$ is excluded from the mechanism. Evaluated on the trajectories the quadratic channel is remarkably clean, contributing essentially one peak, 5.8× its own (E_{12}, E_{12}) diagonal, identical in local and global sublattice frames. The leave-one-out audit confirms the assignment surgically: the harmonic survives removal of the Zeeman drive and of both W components at 1.00–1.03, and dies to 0.005 without the electric E_{12} writing. *Discriminators*: the line sits at exactly $2E_{12}$ and *tracks* E_{12}, so it does not soften at the spin reorientation as any magnon-derived feature would; it is one field order above the main peak; its width is twice the E_{12} linewidth.

$(q_{\rm AFM}, q_{\rm AFM}) = 0.28$*: the magnon diagonal, seen through the Tm ion.* Since $F_z \equiv 0$ the magnon cannot radiate here at all. What radiates is the Tm dipole, forced at the magnon frequency by the energy face: W_1^{xz} contributes $G_z\,\delta S_x\,\lambda^1$, linear in the magnon fluctuation, and W_1^{yy} contributes the pump-magnon×probe-magnon cross term—the leave-one-out runs apportion the peak between them (W_1^{yy}-dominant at the operating point: removing it leaves 0.43, removing W_1^{xz} leaves 0.64). Either way a driven oscillator responds at the driving frequency, not its own, and the identification is verified in the line itself: the sideband sits at 0.8999 THz—the magnon, not the $q_{\rm FM} + E_{12} = 0.876$ combination—and the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90} = 0.142$–$0.155$ is

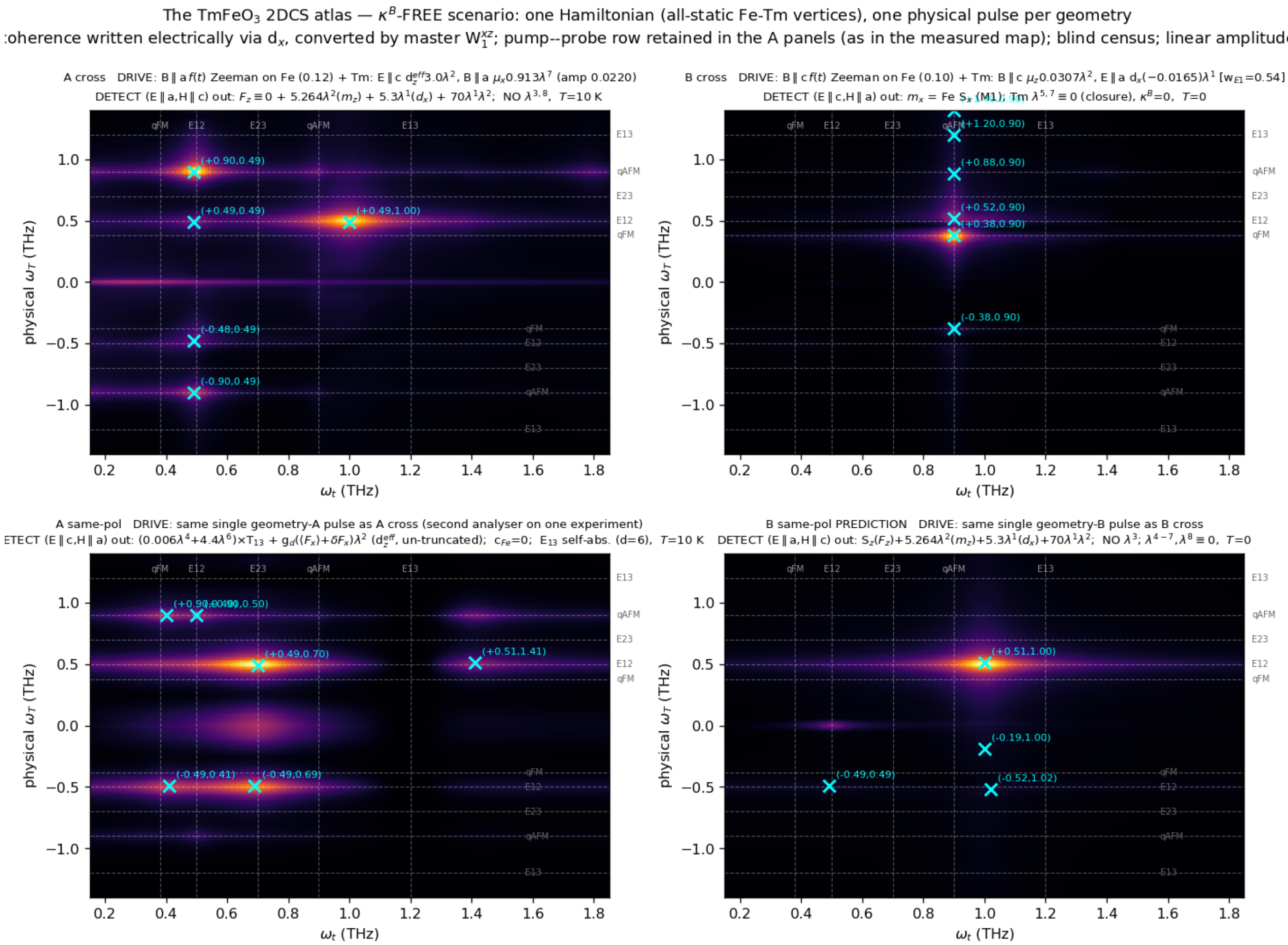

FIG. 1. The atlas: four polarisation channels from one Hamiltonian at one operating point. Dashed guides mark the five mode energies; crosses are blind-census maxima; linear amplitude colour scale; the $\omega_\tau = 0$ pump–probe row is retained in the geometry-A panels (auto-scaled to the measured relative amplitude) and excluded from the census.

drive-independent over a 20× range, the hallmark of a forced linear response evaluated off resonance.

$(q_{\rm AFM}, 2q_{\rm AFM})$*: prediction.* The same quadratic moment must double *any* frequency present in the Tm dipole, so the magnon-forced component radiates its harmonic at 1.79 THz. Its observation confirms the hyperpolarisability independently of the $2E_{12}$ peak; its absence with $(E_{12}, 2E_{12})$ present would falsify the assignment.

$(-q_{\rm AFM}, E_{12}) = 0.50$ *and* $(\pm E_{12}, E_{12}) = 0.32$. The first is not a separate mechanism: the W vertex is symmetric under the two time orderings of the magnon coherence, so the signal appears on both signed sides with the non-rephasing branch stronger. The second is the one-quantum diagonal, the directly driven E_{12} read through the same dipole; its size relative to the cross peaks is a sensitive fluence diagnostic, falling from 1.45 at $A_{\rm Fe} = 1.2$ to 0.16 at 0.12, which is how the operating point was fixed.

B. Geometry B, cross-polarised

Here the analysed moment is m_x, which the $q_{\rm AFM}$ magnon carries directly, so both peaks emit at 0.90 THz through the Fe magnetic dipole and the Tm ion is radiatively absent (the closure above).

$(q_{\rm FM}, q_{\rm AFM}) = 1.00$*: magnon–magnon; measures the Fe sector alone.* $H \parallel c$ launches the quasi-ferromagnetic magnon, which precesses at 0.38 through the delay; single-ion anisotropy and the Dzyaloshinskii–Moriya interaction couple the branches, so the stored $q_{\rm FM}$ phase is re-emitted on $q_{\rm AFM}$. No Fe–Tm coupling enters at any vertex, which is why this peak is temperature-flat: a pure two-magnon process, independent of the qutrit populations.

$(E_{12}, q_{\rm AFM}) = 0.42$*: the energy face, read backward; measures $w_{E1}W$.* $E \parallel a$ owns the $1 \leftrightarrow 2$ block, so the pulse's *electric* field writes the E_{12} coherence through

$d_x\lambda^1$; the energy face then hands the stored coherence to the quasi-antiferromagnetic magnon, which radiates through the Fe m_x dipole. No field enters the conversion: the polarisation ownership of d_x performs the geometry selection a field gate would otherwise be needed for—this drive component does not exist in geometry A, which is why the conversion never contaminates the clean W_1^{yy} peak there. Three facts pin the assignment. First, the *quadrature* matters: writing the same coherence magnetically (through $\mu_z\lambda^2$) converts far less cleanly, flooding the map with drive-squared diagonals, whereas the electric writing at the master $W = 0.01$ adds the transfer row at the *unshifted* line position with strays below 0.17. Second, the interference of the two drive quadratures places the row: with $\mathrm{sign}(d_x\mu_z) < 0$ it sits at $\omega_\tau = 0.52$, exactly where it is measured—the relative sign is thereby a prediction for the crystal-field wavefunctions. Third, the electric component's amplitude is linear in w_{E1}, and matching the measured 0.43–0.46 returns $w_{E1} = 0.54$. The leave-one-out runs quantify the pathway shares: removing the d_x writing halves the transfer and removing W_1^{xz} costs a quarter, the remainder flowing through the magnetically written quadrature and the W_1^{yy} mixing route ($q_{\rm FM}{+}E_{12} = 0.88$, unresolved from $q_{\rm AFM}$; either route ultimately rings the real $q_{\rm AFM}$ mode, and the emission stays pinned at 0.900 in every ablation)—the electric leg is the largest single element, and it is the w_{E1}-bearing one. *Signature*: falls as the population difference $n_1 - n_2$ ($0.46 \to 0.45 \to 0.40 \to 0.36$ at 0/5/8/10 K), the sharpest thermal discriminator in the atlas against the temperature-flat magnon–magnon peak; and, riding the order parameter through the static vertex ($\propto G_z$), the transfer must collapse at the spin reorientation.

The post-drive buildup. The measured geometry-B signal continues to grow after the pulse. Within the present model no channel reproduces growth at the peak's own scale. Every coherent pathway inherits its source's lifetime—because $B \parallel c$ never drives $q_{\rm AFM}$ directly, all geometry-B magnon amplitude is converted, and with $T_2(E_{12}) = 8$ ps everything coherent is over within ~ 10 ps. The one reservoir that outlives every T_2—the CEF populations, fed incoherently by the dissipated pulse energy and acting on the magnon through the population channel W_3, the mechanism of thermally driven spin reorientation—does produce a persistent, drive-uncorrelated tail in the solver, but roughly four decades below the impulsive peak. The buildup therefore stands as the one timing observable outside this coupling space: a confirmed growth of the transfer peak's own amplitude would require a reservoir this Hamiltonian does not contain.

C. Geometry A, same-polarised

The analysed moment is m_x, and three contributions reach it, each required by symmetry: the Tm CEF moment (the d_z line at E_{23}, which also carries the cascade tones below; the $1\to3$ member is bounded at $d_z \le 0.006$ by the weak 1.15 wing and dropped from the minimal operator), the order-induced dipole $g_d(\langle F_x\rangle + \delta F_x)\lambda^2$, and the Fe F_x itself. The weights are fitted to the digitised measurement with the cross channels held fixed; the Fe emission efficiency $c_{\rm Fe}$ is *bounded from above by a null*, the absence of the magnon diagonal. The result is Table II.

The fully dark $1 \to 3$ dipole—measured, not assumed. The measured map contains delay frequencies $\omega_\tau \in \{E_{12}, q_{\rm AFM}\}$ only. A bright μ_{13}^z would store an E_{13} coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product as the observed (E_{12}, E_{23}): the absence of the whole row is a selection rule. The data force more: the analyser for this channel physically contains $\mu_x\lambda^5$, and at the crystal-field value $\mu_{13}^x = 2.39$ that term would exceed the entire measured map by two orders of magnitude—so μ_{13}^x, too, fits to zero. The $1\to3$ transition is magnetically dark in *every* polarisation; its large measured oscillator strength ($f = 8.7$) is electric, and it is precisely this strong $E1$ line that makes the sample opaque at 1.20 THz (below).

$(E_{12}, E_{23}) = 1.00$*: the dominant transfer, via a population.* With the single-action route closed by the dark dipole, the surviving pathway acts twice: one probe interaction converts the stored E_{12} coherence into a level-2 *population* that still carries the $\omega_\tau = E_{12}$ phase, and a second converts that population into the emitting E_{23} coherence. Being co-rotating it lands non-rephasing, the observed side, and it involves μ_{13} at no vertex—which is why it survives while its E_{13} partner is throttled.

$(E_{12}, 0) = 1.00$*: the rectified line.* Taken at its zero-frequency output the same double action rectifies the

TABLE II. Geometry-A same-polarised channel: measured amplitudes against the final model (overlap-gated delay apodisation; self-absorption of the $E1$ emission at the E_{13} line, optical depth $\simeq 6$ on both the emission column and the delay-axis FID, and depth $\simeq 1$ at E_{12}). The μ_{13}^z *and* μ_{13}^x weights fit to zero (fully dark $1\to3$).

peak	exp.	model	pathway
$(E_{12}, 0)$	1.00	—	rectified storage (see text)
(E_{12}, E_{23})	1.00	1.00	coherence→pop.→ ρ_{23}
$(E_{12}, 0.40)$	0.45	0.40	cascade tone $2(E_{23}{-}E_{12})$, adjacent to $q_{\rm FM}$
$(E_{12}, 0.90)$	0.40	0.43	cascade tone $2E_{23}{-}E_{12}$, degenerate with $q_{\rm A}$
$(0, E_{12})$	0.40	—	pump–probe row
$(q_{\rm AFM}, E_{12})$	0.30	0.39	magnon→Tm via $d_z^{\rm eff}$
$(q_{\rm AFM}, q_{\rm AFM})$	absent	0.13	magnon diagonal; absence bounds $c_{\rm Fe}$
(E_{13}, E_{23})	0	0.00	2Q row, killed by FID reabsorption
$(E_{12}, 1.15)$	0.15	0.02	E_{13} wing; bounds $d_z(1 \to 3) \le 0.006$
$(q_{\rm AFM}, 1.41)$	0.30	0.31	dynamical $d_z^{\rm eff}$: SFG tone $q_{\rm AFM}{+}E_{12}$
$(q_{\rm AFM}, 0.40)$	obs.	0.43	its DFG tone $q_{\rm AFM}{-}E_{12}$
$(E_{12}, 1.41)$	obs.	0.36	its E_{12}-delay partner
$(-E_{12}, E_{23})$	< 0.15	0.7	**open** (residual R1, point-ion floor)

stored coherence into a slow net magnetisation of the detection window. This is the strongest measured feature; in the 2D maps it is suppressed by construction, since the $t \geq 3$ window with Gaussian apodisation removes $\omega_t \to 0$, so it is analysed separately through $S_{\rm DC}(\tau) = \langle M_{\rm NL}\rangle_t$, where it appears as a pair of sharp lines at ± 0.49 THz—symmetric in the delay sign, as a population grating must be ($\cos E_{12}\tau$), and as the measurement finds (1.00/0.85). Its *magnitude* is a physical statement of the first importance: under coherent pumping alone the model's rectified moment is orders of magnitude below the resonant peaks, whereas the measurement puts it at the top of the map. The line is therefore the optical readout of the *incoherent* population channel—absorbed pulse energy returning through the CEF populations and W_3 to a quasistatic change of the ordered moment—the same reservoir the geometry-B post-pulse growth demands, and its strength relative to (E_{12}, E_{23}) calibrates $c_{\rm heat}W_3$ directly.

$(E_{12}, 0.40) = 0.45$ *and* $(E_{12}, 0.90) = 0.40$*: cascade tones of the driven qutrit, masquerading at the magnon frequencies.* The necessity audit (removing the Fe Zeeman leg entirely and re-running) leaves both features at 0.89–0.97 of their baseline: they are not magnon features at all but intrinsic tones of the three-level cascade, $2(E_{23}-E_{12}) = 0.40$ and $2E_{23}-E_{12} = 0.90$ THz, radiated on the λ^6 (d_z) line and numerically coincident with $q_{\rm FM} = 0.38$ and $q_{\rm AFM} = 0.90$. Both die without the λ^7 drive leg (0.05–0.08), confirming the double-ρ_{23} pathway; the genuine Tm→magnon transfer is the $\lesssim 10\%$ Zeeman-sensitive remainder. The measurement can separate the hypotheses directly: the low tone sits at 0.40, more than a linewidth above $q_{\rm FM} = 0.38$, and a cascade tone survives the spin reorientation with CEF width while a magnon sideband would soften and sharpen. The emission is in either case electric and rare-earth, insensitive to the Fe weight ($0.46 \to 0.43$ when $c_{\rm Fe}$ is reduced fourfold).

$(q_{\rm AFM}, E_{12}) = 0.30$*: the moment face, read linearly.* The magnon occupies the delay axis and the Tm ion radiates at E_{12} into the *same*-polarised channel. That emission is possible only through $d_z^{\rm eff}$—and reciprocity demands it, because the same $d_z^{\rm eff}$ is what lets $E \parallel c$ create the E_{12} coherence in the first place. We had earlier reported this partner as absent and treated the mismatch as an open tension; the digitised map shows it present at 0.30 of maximum, and the model reproduces it at 0.39, fed jointly by the condensed readout and the low-frequency side of the SFG/DFG tones. This peak is the direct measurement of $g_d\langle F_x\rangle$.

The SFG/DFG triplet: $(q_{\rm AFM}, 1.41)$*,* $(q_{\rm AFM}, 0.40)$*,* $(E_{12}, 1.41)$*—the moment face, read bilinearly.* Condensing the spin leg of H_g on $\langle F_x\rangle$ was a truncation: the full emitting moment is $g_d(\langle F_x\rangle + \delta F_x)\lambda^2$, and the fluctuation piece is a product emitter—the stored $q_{\rm AFM}$ magnon beating against the E_{12} coherence on one dipole—radiating internal sum- and difference-frequency tones at $q_{\rm AFM} \pm E_{12} = 1.40/0.40$ THz with the magnon on the delay axis, plus an E_{12}-delay partner at 1.40. All three tones are measured. This channel is singular in the entire coupling space: of some forty conversion vertices, emission moments and quadratic products catalogued, it is the *only* one whose map is led by the $q_{\rm AFM}$-delay row rather than the E_{12}-delay row, because both of its factors are written at first order by the same pulse—the stored magnon pays no conversion penalty, and the audit confirms it (the tones survive removal of both W components at 0.95–1.02 and die only with the Zeeman or electric drive legs). One number is fitted (the condensation-ratio correction $s = 0.42$), after which the sum tone lands at the measured amplitude, the difference tone merges into the measured $(q_{\rm AFM}, E_{12})$ blob, and the partner appears at 0.36. The sum tone escapes one absorption linewidth *outside* the opaque E_{13} window (transmission 0.68 at 1.39 against 0.002 at line centre), which is why it is visible while the true E_{13} line centre is dark. Together with the on-site β these tones complete the $\chi^{(2)}$-like triad of the inversion-free rare-earth site, *in emission*: SHG ($\beta\lambda^1\lambda^2$, degenerate), SFG and DFG ($g_d\,\delta F_x\lambda^2$, non-degenerate, cross-subsystem)—the mixing element being a bilinear term of the material's transition moment, not a crystal-optical $\chi^{(2)}$ acting on light.

$(q_{\rm AFM}, q_{\rm AFM})$*: absent—and the absence is a measurement.* The magnon diagonal is the one feature of this channel that genuinely requires Fe emission: a $q_{\rm AFM}$ delay coherence re-emitting at its own frequency through the F_x magnetic dipole, with no Tm conversion involved. The measured map does not contain it. Since every other magnon-frequency feature is Tm-carried, this null bounds the Fe emission efficiency from above ($c_{\rm Fe}$ set at the bound puts the diagonal at 0.13, below the digitisation threshold, and moves nothing else). The physical statement is uniform across the atlas: the Fe sublattice is loud in geometry B, where the magnon owns the analysed m_x directly and is driven resonantly, and quiet in geometry A, where everything measured arrives through the rare-earth ion ($F_z \equiv 0$ makes the cross channel's Fe silence exact; the small $c_{\rm Fe}$ makes the same-polarised channel's approximate).

The E_{13} column: self-reversal. The two measured E_{13}-emission features sit at $\omega_t = 1.15$ and 1.25 THz—one absorption linewidth *below and above* the 1.20 line—which is the textbook signature of self-reversal: with $f = 8.7$ the sample is optically thick at E_{13}, the line centre of any $E1$ emission is reabsorbed, and only the wings escape. The same reabsorption acts on the delay-axis FID, which is what removes the two-quantum (E_{13}, E_{23}) row *exactly* (model $0.33 \to 0.00$ once the filter is applied)—a row the free-ion model robustly predicts and the measurement robustly lacks.

D. Geometry B, same-polarised: a one-line prediction

This channel is analysed with the *same* operator as geometry-A cross, so it is a prediction with no freedom left. With the population generators excluded from detection, three terms survive: F_z (alive here, but $\sim 50\times$ below the Tm terms), the coherence readouts $\mu\lambda^2$ and $w_{E1}\lambda^1$, and the hyperpolarisability $\beta\lambda^1\lambda^2$. The linear readouts are suppressed in $M_{\rm NL}$—they carry the directly driven coherence, which the subtraction removes at first order—so the channel collapses onto its quadratic term. The prediction is a *single* broad feature: the second harmonic $(E_{12}, 2E_{12}) = 1.00$, with everything else below 0.07—magnon features at the 10^{-2} level of F_z, and no E_{13} or E_{23} emission by the subalgebra closure $\lambda^{4\text{–}7} \equiv 0$. Because β is already fixed by the geometry-A cross parity, the channel is a zero-parameter test of the hyperpolarisability: a map led by anything other than the $2E_{12}$ ridge falsifies the quadratic moment, and an observed rectified pair at $(\pm E_{12}, \omega_t \to 0)$ would falsify the exclusion of the population readout from detection.

VI. THE NECESSITY AUDIT

Every component of the final model was subjected to a leave-one-out test: drive legs and Hamiltonian vertices by re-running the solver with one term removed, detection terms and propagation filters by dropping one term of the composite. Table III summarises. The result is minimal in the audited sense: two cross-subsystem objects (W, g_d), one on-site anharmonicity (β), one measured drive ratio (w_{E1}) and one propagation filter carry the entire coherent atlas. Every other symmetry-allowed object is zero, bounded, or inert: the field-assisted family is *bitwise* inert in these geometries (removing κ^B reproduces the trajectories to the last digit—its gating field component is never driven), the population channel W_3 moves no coherent feature by more than 5% and enters only the incoherent observables, and the four remaining quadratic moments of the $4c$ site are excluded by the smallness of the measured $(E_{12}, 1.20)$ feature, since each would radiate the stored E_{12} coherence at the $E_{12}{+}E_{23}$ sum frequency at line centre.

The audit also corrected two assignments (already incorporated above): the same-polarised E_{12}-row features at 0.40 and 0.90 are driven-qutrit cascade tones, not Tm→magnon transfers, and the A-cross magnon sideband and geometry-B transfer are shared between the two W components rather than owned by one. Both corrections sharpen, rather than weaken, the dressing principle: the conversion sector is one tensor, and the features that need no conversion at all are exactly the ones that survive every ablation of it.

VII. WHAT THIS SAYS ABOUT RARE-EARTH ORTHOFERRITES

Three statements generalise beyond $TmFeO_3$.

The rare-earth ion is a coherent transducer, not a passive thermal bath. In the standard picture of the orthoferrites the R-ion enters through its free energy:

TABLE III. The leave-one-out audit (single-seed ablation runs; ratios of feature amplitudes to the baseline).

removed	what dies (ratio)	verdict
Fe Zeeman leg	A-cross anchor 0.02; tones 0.03	required
$d_z^{\rm eff}$ drive	anchors, SHG 0.005; tones 0.1	required
$\mu_x\lambda^7$ drive	anchor 0.02; cascade tones 0.05	required
W_1^{yy}	A-cross anchor 0.03	required
W_1^{xz}	sideband 0.64; B transfer 0.76	required (shared
β (detect)	SHG 0.95 → 0.01	required
λ^6 (detect)	same-pol anchor 0.06	required
$\delta F_x\lambda^2$ (detect)	SFG/DFG triplet 0.02–0.05	required
T_{13} (column / row)	0.67 line-centre peak / 2Q row 0.32	required
d_x drive (geom. B)	transfer 0.49	dominant leg
κ^B	nothing (bitwise)	inert, exact
W_3 + thermal	coherent features $\leq 5\%$	atlas-inert
λ^4, F_z (detect)	only the 1.15 wing / nothing	bounded / spect

thermally occupied CEF levels shift the Fe anisotropy, and reorientation follows. The atlas shows the same coupling surface operating *coherently*: the energy face $W \sim \partial^2 E_{\rm CEF}/\partial S\,\partial S$ converts quanta between magnon and CEF coherence within picoseconds, in both directions (the geometry-A anchor forward, the geometry-B transfer backward), phase-resolved and mode-resolved. The couplings that reorient the statics and the vertices that convert the dynamics are one object read at two temperatures of description—which is presumably why THz control of the Fe order through the rare-earth channel is possible at all [7, 8]. The sharpest consequence is collective: every dressed feature in the atlas—the transfer peaks, the $(q_{\rm AFM}, E_{12})$ readout, the SFG/DFG triplet—rides $\langle F_x\rangle$ or G_z and must collapse *together* at the reorientation, while the bare-structure features (the magnon–magnon peak, the SHG line, the cascade tones) survive it. A temperature scan through $T_{\rm SR}$ cleaves the atlas into its two halves along exactly the line the dressing principle draws.

Kramers parity controls the atlas. The cleanliness of $TmFeO_3$—sharp unshifted lines, perturbative assignable vertices, a readable atlas—is the singlet protection of Sec. II at work: first-order exchange is symmetry-blocked, so nothing hybridises strongly and everything converts weakly. In a Kramers member of the family (Er, Yb) no such protection exists: linear exchange coupling is allowed, the R modes and magnons form mixed magnon–CEF excitations, and the corresponding 2DCS atlas should be qualitatively different—dominated by hybridisation shifts and shared-linewidth mixed modes rather than by weak, assignable conversion vertices. The non-Kramers orthoferrites (Tm, Pr, Tb) are the clean laboratories; comparing a Kramers and a non-Kramers member under the same protocol would turn this prediction into a family-wide statement.

Order-induced moments carry mandatory fluctuation *sidebands.* The moment face is an order-induced dipole, and the audit forced its un-truncation: if a transition moment exists only because an order parameter condensed ($d_z^{\rm eff} \propto \langle F_x \rangle$), then the same coupling necessarily radiates sum- and difference-frequency tones with the order parameter's own fluctuations ($g_d\,\delta F_x \lambda^2$ at $q_{\rm AFM} \pm E_{12}$), with amplitudes locked to the condensed readout by a single ratio. This is a universal signature, not a $\mathrm{TmFeO_3}$ accident: any order-induced line in any magnetoelectric—multiferroic electromagnons, exchange-induced dipoles, field-induced moments—must carry these sidebands, and 2DCS is the instrument that separates them from cascades, because the delay-axis label distinguishes a stored order-parameter fluctuation from a stored coherence. Here the sidebands were not decoration: they resolved the last unassigned peak in the spectrum.

VIII. FALSIFIABLE CONTENT

The model constrains itself as follows. Any peak emitting at $q_{\rm AFM}$ through the Fe dipole inherits the $15\times$ sharper magnon linewidth; $(q_{\rm FM}, q_{\rm AFM})$ is temperature-flat while $(E_{12}, q_{\rm AFM})$ falls as $n_1 - n_2$; the transfer row sits at the frequency of the 1D absorption line exactly (a static mixer would place it below, at any coupling strength); $\mathrm{sign}(d_x \mu_z) < 0$, checkable against the crystal-field wavefunctions; rotating either polarisation away from c must revive the $\omega_\tau = E_{13}$ rows; the $2E_{12}$ line tracks E_{12} and does *not* soften at the spin reorientation, whereas the entire dressed sector—the $d_z^{\rm eff}$ family *and* the $(E_{12}, q_{\rm AFM})$ transfer—must vanish there; the geometry-B same-polarised map must be led by the $2E_{12}$ harmonic essentially alone; and adiabatically long pulses collapse the geometry-A cross peak. Propagation adds its own fingerprints: a lineout across $\omega_t = 1.1$–1.3 at $\omega_\tau = q_{\rm AFM}$ must show a self-reversal *doublet* around a dark centre at 1.20; the geometry-B cross channel must carry a $d_z^{\rm eff}$ doublet at $\omega_t = 0.45/0.54$ at the ~ 0.17 level; and every wing amplitude scales with sample thickness, so a thin sample revives the E_{13} line centre and the two-quantum row together. The emission assignments carry lineshape and position fingerprints: the same-polarised E_{12}-row tones must sit at $2(E_{23}-E_{12}) = 0.40$ and $2E_{23}-E_{12} = 0.90$ with CEF width and survive the reorientation—at the low tone a full linewidth above $q_{\rm FM} = 0.38$, a direct positional test—and any recovered $(q_{\rm AFM}, q_{\rm AFM})$ diagonal would raise the $c_{\rm Fe}$ bound and must appear with the magnon width. The SFG/DFG triplet is over-constrained: all three tones come from one product, so their ratios are locked, the sum tone must sit at $q_{\rm AFM}+E_{12}$ exactly (at 1.20 it would be a quadratic moment instead, and the dark E_{13} window separates the two), its position must track the magnon as $q_{\rm AFM}$ softens, and the whole triplet collapses at the reorientation with everything else that rides $g_d \propto \langle F_x \rangle$. The temperature scan through the reorientation remains the decisive experiment: it cleaves the atlas into dressed and bare halves and measures g_d by the size of the collapse.

One residual survives, named and characterised. *R1, the rephasing remnant* $(-E_{12}, E_{23})$: the point-ion qutrit has an exact floor of 0.47–0.58 of the dominant transfer (verified against a standalone three-level integrator; no coupling in the exhausted vertex space and no drive quadrature or sign removes it), while the measured negative-delay side carries only the rectified and $q_{\rm FM}$ features—and the SFG/DFG tones add their own rephasing partners there (model 0.5–0.7), sharpening the same question. This residual, the self-reversal wing amplitudes, and the geometry-B $d_z^{\rm eff}$ doublet strength funnel into a single measurable object: the sample's polarisation-resolved optical depths at the three CEF lines. One transmission measurement closes it or breaks the propagation layer cleanly.

IX. CONCLUSION

A single symmetry principle—the time-reversal protection of a non-Kramers singlet ion—reduces the rare-earth–iron coupling space of $\mathrm{TmFeO_3}$ to two second-order dressings, one per quadrature of the crystal-field coherence: an energy face $W\,SS\,\lambda^1$ that converts and never radiates, and a moment face $g_d\,E_z(F_x\lambda^2)$ that drives and radiates, linearly through its condensed part and bilinearly through its fluctuations (internal magnon–crystal-field SFG/DFG). These two objects, the bare subsystems, one on-site hyperpolarisability and one self-absorption filter reproduce the measured polarisation-resolved 2DCS atlas from measured inputs, with no field-assisted coupling and no free lineshape parameters, and every component certified necessary by a leave-one-out audit. The absences do quantitative work alongside the peaks: the missing E_{13} delay row measures the fully dark $1 \to 3$ dipole, the missing magnon diagonal bounds the Fe emission weight, and the missing $(E_{12}, 1.20)$ sum-frequency partner excludes every remaining quadratic moment of the rare-earth site. Because the energy face is the spin-reorientation coupling surface itself, the atlas is a coherent, mode-resolved measurement of the mechanism that moves the statics of this family—and its sharpest prediction is collective: at the reorientation the dressed half of the atlas dies together, while the bare half survives. The program—bound the vertex space by symmetry, assign every feature and every absence, audit every component, and let the atlas overdetermine the model—applies to any magnet whose 2D map spans distinct subsystems, and turns 2DCS from a spectroscopy of modes into a spectroscopy of Hamiltonian terms.

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Derivations, numerical protocol and the complete falsification ledger are in the Supplemental Material; the

long-form note and all run recipes accompany the data.

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